

Automation of image-based measurement of battery cell features by computed tomography and synthetic training data

Daniel Evans^a, Simon Beckmann^{b,c}, Kevin Talits^d, Claas Tebruegge^b, Julia Kowal^a

^a Electrical Energy Storage Technology, Technical University Berlin, Berlin, Germany

^b HELLA GmbH & Co. KGaA, Lippstadt, Germany

^c Business Information Systems, esp. AI-based Systems, University of Potsdam, Potsdam, Germany

^d Chair of Energy Conversion, TU Dortmund University, Dortmund, Germany

ARTICLE INFO

Keywords:

Automated battery inspection
Battery cell production
Computed tomography
Computer vision
Li-ion batteries

ABSTRACT

Due to process variations in the production of lithium-ion batteries (LIBs), cells of one production batch can show a variation in physical features, inhomogeneities, and defects. These can impact the performance and safety of the cells and should be detected, and if accepted in tolerances should be measured accurately. The cell features are often unknown to manufacturers of battery modules and packs. Hence, computed tomography (CT) imaging could provide insight into the cells' quality, allowing the measurement of relevant battery cell features. However, the high number of cells requires an automation of cell inspection. This work focuses on the challenge of automated image processing and provides an image-based workflow measuring multiple cell features based on a single CT scan. Both classical computer vision (CV) and machine learning (ML)-based image algorithms are applied within the developed workflow. To train, test, and validate the convolutional neural network (CNN)-based algorithms, artificially generated training data is created and used due to the scarcity of training data, which can form a bottleneck in CNN-model development and evaluation. Hence, the generation of synthetic training data shown in this work can reduce the need for costly laboratory CT scans before adoption in serial production environments. The results show the promising potential of synthetic training data and the automated approaches to measure cell features, specifically the electrodes' windings, the corresponding length and width, as well as the anode overhang.

Introduction

Process variations in the production of lithium-ion battery cells can lead to inhomogeneities and defects. Some inhomogeneities and defects must be avoided due to their substantial effect on cell performance and safety. However, other inhomogeneities and variations may be acceptable in tolerances. As a result, cells of one cell type often show a distribution of electrical properties [1–11]. The cell supplier can carry out improvement and direct supervision of cell production processes and utilize in-line production data for optimization [12–14]. In contrast, module and pack producers often depend on additional cell testing to ensure the quality of the cells.

Non-destructive evaluation of lithium-ion battery cells can be performed by a comprehensive set of electrical and non-electrical methods as shown by [10,15]. Among these, X-ray-based CT shows a high potential for localization and detection of inhomogeneities in battery cells [15,16]. Based on multiple X-ray projections, CT is a non-destructive measurement method, enabling additional insight into the manufacturing quality of the cell. Visualization, detection, and localization of

inhomogeneities and defects are possible on several levels, including electrode and cell levels [17]. In the literature, several examples are given for the visualization via CT imaging, including the measurement of anode and cathode size, anode overhang, foreign matter contamination (FMD), anode–cathode and jelly roll misalignment, number of windings, tab burrs, electrode gapping, non-uniform electrode coatings and thickness, and uncoated regions, delamination as well as cuts and scratches [10,15–28].

This work focuses on the CT-based feature measurement in cylindrical cells, specifically their electrode length, corresponding windings, and width, as well as the anode overhang. The anode overhang results from both the electrode dimensions and the relative position of the anode and cathode. Typically, the anode is designed slightly larger than the cathode, creating a geometric overhang intended to prevent lithium plating at the anode edges [29]. Additionally, the electrode dimensions determine the cell's electrochemical active area. Monitoring these cell features in production is crucial because consistent electrode size and

* Corresponding author at: Electrical Energy Storage Technology, Technical University Berlin, Berlin, Germany.
E-mail address: daniel.b.evans@campus.tu-berlin.de (D. Evans).

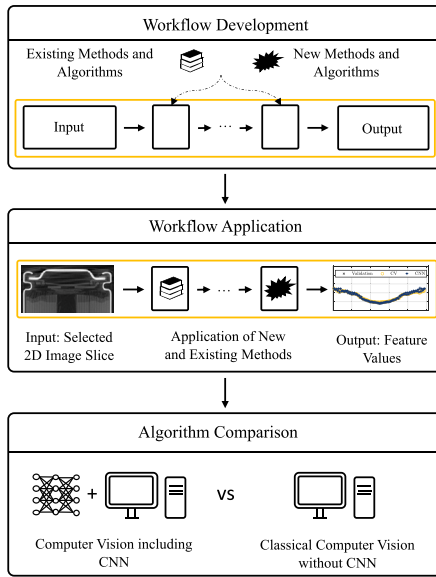


Fig. 1. Methodical approach for the study comprising workflow development for automation of image-based measurements by CT, workflow application on selected CT slices, and comparison of CNN-based and CV-based algorithms without ML.

overhang are key to ensuring battery performance and safety. Once the jelly roll is inserted into the battery cell case, X-ray-based methods currently offer the only opportunity to assess these features. Consistent measurement reduces the response time for correcting deviations in size or overhang, thereby lowering the risk of rejects. Early trend detection allows for the implementation of countermeasures before deviations exceed predefined tolerances.

However, there are several challenges for the automated application of 3D CT in serial production of lithium-ion batteries (LIBs) as part of inline measurements, which refer to measurements on the production shopfloor and integrated directly in the production line [30]. Inline measurement approaches must comply with short cycle times. Therefore, automated probe handling, fast scanning times (limited by integration/exposure time) in combination with efficient filtering and volume reconstruction, and automated workflows for image and voxel cloud evaluation are required [31] to enable the application of CT for high numbers of produced LIBs.

An X-ray-based alternative, 2D X-ray, has been demonstrated for several use cases in battery cell production, including the detection of FMD and anode-cathode misalignment, and provides significant benefits regarding imaging time and complexity of material handling [16, 20]. However, advancements in CT scanning time and high-speed automated handling [28,32] and the capability to extract additional product features highlight the potential of CT applications in future serial production environments. When sample testing is conducted in a laboratory [30] or through offline measurements (sometimes referred to as atline) on the shopfloor [30], rather than as inline measurements directly within the production line, manual probe handling and scanning time are not as crucial. However manual inspection of the cell, without usage of automated inspection workflows and feature extraction can be very time-consuming and may not be able to extract all cell features, inhomogeneities, and defects, e.g., due to additional calculations necessary. This work therefore focuses on the post-processing of the CT images, the automation of inhomogeneity assessment, and the measurement of physical product features that impact the cell's electrical properties. A focus of the investigation is hereby laid on the general feasibility of the methods to assess and measure the cell's features accurately while finetuning of the image processing step and analysis time is not in the scope of this work.

Fig. 1 shows the methodical approach in this work. At first, the developed workflow and relevant steps are presented. While the measurement of features, such as electrode length measurement, has been described in the literature, both an overarching workflow for automation of cell inspection via CT, as well as specific CV and CNN based-algorithms have not yet been presented, assessed and compared for the usage on cell level CT in the literature. This work aims to fill this research gap.

The workflow is based on CT-based approaches for non-destructive-testing (NDT) and state-of-the-art methods for image recognition and analysis, as the actual analysis of the battery cell geometry is performed on selected 2D CT slices. The method consists of multiple steps, from image pre-processing to algorithm application for inhomogeneity assessment and feature extraction. It reflects recent literature on advancements in feature extraction. Significant progress has been achieved in feature extraction, particularly in analyzing CT scans, thanks to advancements in ML. These advancements are attributed to machine learning models, which facilitate the automated extraction of features from CT scans [33]. This work enables the application of new models and transfer of existing models such as you only look once (YOLO)v5 for CT based measurements by providing both a framework for application and novel methods for training data generation. The advancements and recent literature are summarized, and novel methods are introduced. In addition, this work provides innovative ways to generate training data that is usually scarce during the development of inspection methods based on CT and can form a bottleneck for CNN training and evaluation. These also reduce the need for costly laboratory CT scans before adoption in serial production environments.

Secondly, the approach is carried out on a cylindrical cell. A CT based study on a cylindrical cell is performed. Features of a cylindrical cell are measured by automated execution of the workflow. The electrodes size, specifically the length, windings and width are measured, in addition to the anode overhang.

The application of the workflow is described for both traditional CV algorithms and CNN based algorithms. In this work, the term CV refers to CV that relies on theoretical and mathematical models in contrast to ML models, particularly those employing convolutional neural networks (CNNs). Depending on the measured feature and the underlying feature extraction algorithms, selected steps of the workflow are applied.

In the third step, the measurement results of both algorithm categories (with and without the usage of ML algorithms, including CNNs) are compared with the help of validation measurements. These are carried out manually via CT imaging. Benefits and challenges for the approaches and their application for CT based measurements are examined.

Methods

Due to the nature of the battery cell assembly (multiple, partly flexible components), this work offers an alternative to often-used CT-based approaches for metrology. Conventional methods generate computer-aided design (CAD) volume or surface models via segmentation and surface determination directly from the 3D CT voxel cloud. The resulting CAD model is then compared with the original CAD file of the component or directly used for measurements [34–36].

In contrast, the approach in this work is based on processing 2D CT image slices. Although previous research has demonstrated the use of CT image slices for analysis [21–23,37,38], our work refines these methods by providing a structured workflow for image analysis. In the following sections, we present an overview of the workflow, then summarize previous work for each processing step. Finally we introduce novel methods to the field of CT imaging for lithium ion batteries, that enable the generation of synthetic imaging data for lithium-ion battery cells.

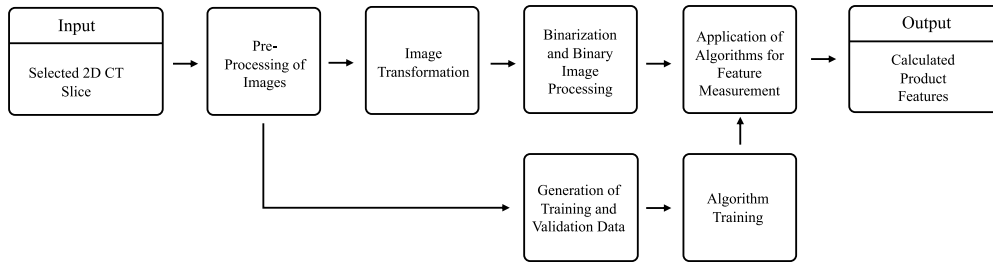


Fig. 2. CT image processing workflow for battery cell feature measurement and extraction from input to output. Selected 2D CT image slices are processed and calculated product features are given as output of the workflow.

Table 1

Overview of features, including an assigned feature class and the affected or contributing LIB component. In addition performed steps of the workflow are summarized to assess the examined feature.

Feature class	Component	Feature	Image pre-processing	Binarizing/Processing	Image transform	Algorithm training	Algorithm application
Radial CT slice	Anode, Cathode	Length, Windings	CV, CNN	CV	CV	CNN	CV, CNN
Axial CT slice	Anode	Overhang	CV, CNN	CV		CNN	CV, CNN
	Anode	Width	CV, CNN	CV		CNN	CV, CNN
	Cathode	Width	CV, CNN	CV		CNN	CV, CNN

Automated workflow for computed tomography based feature measurement

The usage of 2D CT image slices enables the application of widespread image-based algorithms for segmentation, detection and consequent feature measurement inside the battery cell. In this study, the workflow is applied to quantify anode overhang, as well as anode and cathode dimensions, which are crucial to cell functionality. Fig. 2 gives an overview of the proposed workflow. Depending on the feature of interest, 2D CT image slices are selected and pre-processed. Subsequent steps include image transformation, binarization, and processing of the binary images.

The cell features are computed using algorithms applied to the original, pre-processed, transformed, or binary images. The following sub-chapters provide detailed descriptions of each processing step and recent advancements. Not all steps are required for every measurement task. For example, CNNs may be applied directly to pre-processed images. Table 1 gives an overview by highlighting the steps used to measure each cell feature in our work. The workflow is required to be flexible, allowing for adaptation to various scan types and features, facilitating the comparison of different approaches.

Input: Volume data generation, image stack extraction and image selection

The acquisition of 2D CT slices is essential for the following image processing step. At first, CT volume data is generated by reconstructing a 3D voxel volume from 2D X-ray projections. The voxel resolution and image quality are determined by various factors, including probe properties (e.g., size, geometry, density, and material absorption coefficients), CT scanner characteristics (e.g., radiation source and detector type), and scanning parameters (e.g., tube current, tube voltage, filters, component wear, region of interest, probe positioning and orientation, integration time, and number of projections) [31,34,36,39–41]. To improve the resolution of a specific region of interest, selective or composite scans can be used [34]. In this work a composite scan was used, allowing a higher scan resolution.

To extract the image after the scan, single or multiple 2D CT image slices can be generated based on the 3D voxel volume. Generation is possible along a predetermined axis at specified distance intervals or by rotating around a fixed axis in defined angular increments. Based on the 2D image stack and the features of interest, representative CT slices are selected. For cylindrical cells two CT-image slice types are typically of interest, the radial and the axial CT image slice. These are shown in Fig. 3 for the cell used in this study.

Pre-processing of image(s)

Pre-processing incorporates operations such as resizing the image, enhancing contrast, blurring, filtering, or sharpening the images depending on the feature of interest [42]. For CT images an important pre-processing step usually includes de-noising, using filters such as the average, median, Wiener, and Gaussian filters. Additionally, image enhancement is facilitated through techniques like histogram equalization or direct contrast mapping via both linear and nonlinear functions [43]. In the field of battery cells, for the extraction of windings of spirally wound cells, an adaptive edge enhancement algorithm based on a linear small kernel median filter, an unsharp mask, and a Gaussian difference filter has been deployed on CT slices of cylindrical battery cells to prepare the extraction of the boundary between anode and cathode [21]. In addition approaches have been shown that only examine specific regions of the battery cell or image information along selected contours [22,23,37,38]. While more sophisticated methods have already been presented for CT imaging, the focus of this study lies on high quality images of cylindrical cells that require only minor pre-processing for application of CNN and CV-based applications.

Image transformation

For cylindrical cells, applying a polar transformation to radial CT image slices can facilitate subsequent image processing. In the radial view, the anode and cathode appear as spiral structures (Fig. 3(b)), complicating the length estimation of each winding. Several studies [21,44] have demonstrated that converting images from cartesian to polar coordinates straightens each winding, thereby facilitating their separation and enabling the extraction of features such as winding angle and count using classical CV methods without utilization of machine learning. This transformation typically not necessary for axial CT slices, where the anode and cathode already appear as straight lines (Fig. 3(c)).

Image binarization and binary image processing

Due to differences in X-ray attenuation and resulting gray values, materials allow for their separation if the resolution is high enough. Binarization is a step in image processing that converts an image a grayscale image into a binary image. It thereby facilitates the separation of components in the image while preserving the structural details.

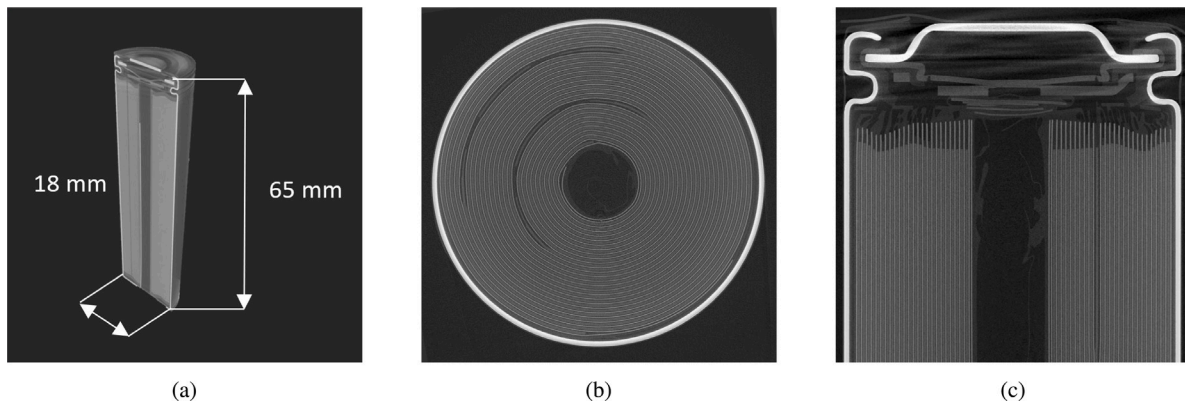


Fig. 3. CT scan of cylindrical cell in used in this study. (a) shows the dimensions of the scanned battery cell. (b) shows a radial and (c) shows an axial CT slice.

In our work binarization is applied to identify anode and cathode areas as part of CV-based methods allowing subsequent measurement of anode and cathode width, length and overhang. The resulting images contain only black-and-white information [45] where a distinct component is represented as white against a black background.

Several methods can be applied for selecting a threshold, such as global thresholds based on a single threshold for the entire image and local methods that include separating the image and finding the threshold for each sub-image [46]. The processing of binary images involves several operations and can be performed, e.g., to remove specific geometries based on their size, such as artifacts within the binary image. Examples are morphological operations, distance transforms, and finding connected elements [45]. Morphological operations can include closing, opening, erosion, and dilation. Finding connected elements and ranking the components, e.g., by area, to identify and delete the largest component from the image can be valuable.

However, while binarization efficiently separates structures, it also reduces the image information and may neglect subtle gray-value variations. Consequently, if the measurement task requires the detection of fine variations, such as slight density differences within a material, relying on binarization might result in a loss of critical information. This is also why we do not apply binarization before implementation of ML-based methods in our work — as also highlighted in Table 1.

Application and training of algorithms for feature measurements

Several algorithms can be applied to the measurement of the features. Based on the initial feature measurements, further calculations can be applied to extract secondary information, such as the statistical distribution of specific features and inhomogeneities, such as anode overhang. In this work, classical CV methods and CNN driven methods are assessed separately. CNN methods tend to provide more flexibility and often achieve greater accuracy in image classification and object detection [47]. In contrast, classical approaches can be a viable alternative for less complex tasks.

Traditional CV driven methods often rely on algorithms such as feature extraction of small features such as edge detection, corner detection, or threshold segmentation [47]. The canny and Sobel algorithms are examples of edge detection algorithms [48]. In the realm of X-ray-based inspection of LIBs and process supervision [20], edge detection via the canny algorithm has been used to extract contours of bubbles during the filling step of LIBs. The digital disassembly by [21] is based on contour extraction via a marching cubes algorithm.

In contrast, CNNs draw inspiration from the structure of the brain's neural network. Operating as a specialized variant of feed-forward artificial neural networks (ANNs), CNNs feature locally connected neurons. These local connections are established through a series of filters initialized arbitrarily. Consequently, the primary computational step in

a CNN involves convolving these filters with an input image. The architecture comprises multiple stages for feature extraction, with each stage primarily involving three fundamental operations: convolution, activation of non-linear neurons, and pooling of features [49,50]. In the field of CT scan image analysis and/or LIBs, CNN architectures have found utility in both regression [51] and classification [52] applications. In regression, the objective centers on predicting continuous variables, whereas classification tasks involve the segmentation of images into discrete classes.

Novel approaches for generation of training data and validation data

A large and diverse training data set is an important cornerstone for a robust and generalized artificial neural network (ANN). Therefore image pre-processing is also used for training of CNNs. Machine learning models necessitate vast amounts of labeled data. As CT scans are time-consuming, data augmentation methods can be employed to augment the dataset of CT images, which mitigates overfitting and enhances performance [53]. Additionally, normalization or standardization steps can be implemented to improve the model's accuracy [54].

In situations with little data, more must be collected or produced synthetically. Many problems associated with artificial intelligence (AI) can be traced back to insufficient data, but synthetic generation is an increasingly popular solution to this problem [55]. Using generative adversarial networks (GANs) [56] to learn from real data and enhance the data set by creating synthetic sets is becoming increasingly prevalent. In more simplistic cases and with fewer real images, advanced manipulation techniques and CV automation can be utilized to create more data. As demonstrated by [57], image generation is also possible for CT projections. In addition, the generation of training sets for underrepresented defect classes [58] has been investigated. However, no approaches in the literature are demonstrated to generate training data for battery cells.

The scarcity of CT imaging data makes it difficult to train and evaluate ML-based algorithms. These require a high number of training and validation images. Therefore, algorithms are developed in this work to generate synthetic CT slices for both axial and radial CT slices. For the axial CT slice, a pixel shifting algorithm is introduced to generate synthetic CT image slices. Specifically, the algorithm aims at the generation of new images and automated labeling. The approach is shown in Fig. 4. At first, a single high-resolution CT image slice is extracted from an CT scan. The initial image slice is labeled and cropped manually to increase the label area in comparison to the overall image area. Randomized shortening of the anode and cathode is performed. Residual areas are filled, and a Gaussian fade is applied to the newly formed edges around the anode and cathode. A new label is calculated for the new tips of the anode and cathode.

The method can be transferred to the radial view of the cell. However, for applying the pixelshift algorithm, the original image

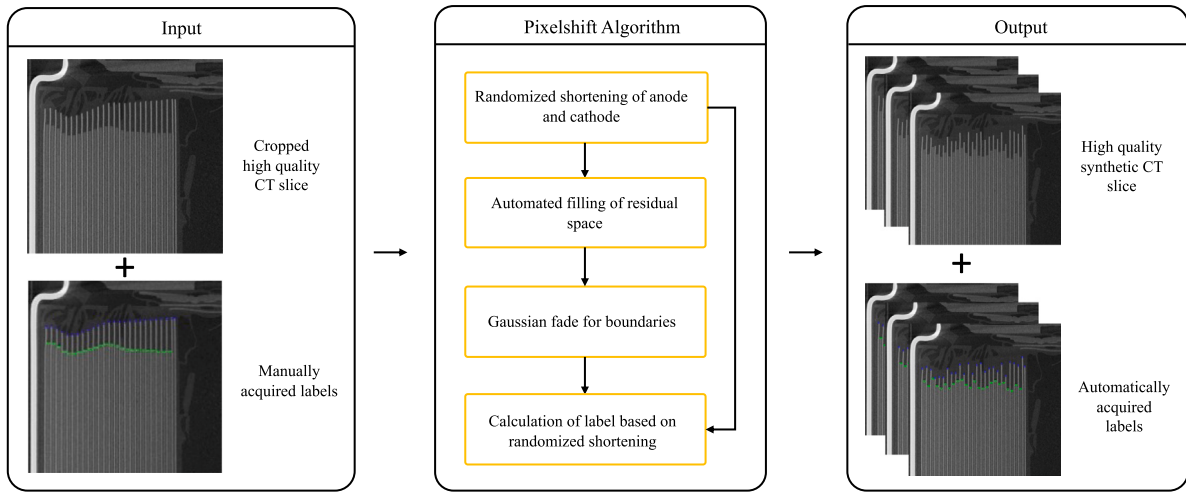


Fig. 4. CT image slice generation by pixelshift method for training of CNN based measurement algorithms of axial CT slice. High resolution CT slices and manual labeling serve as input to generate synthetic CT slices and automate the process of labeling.

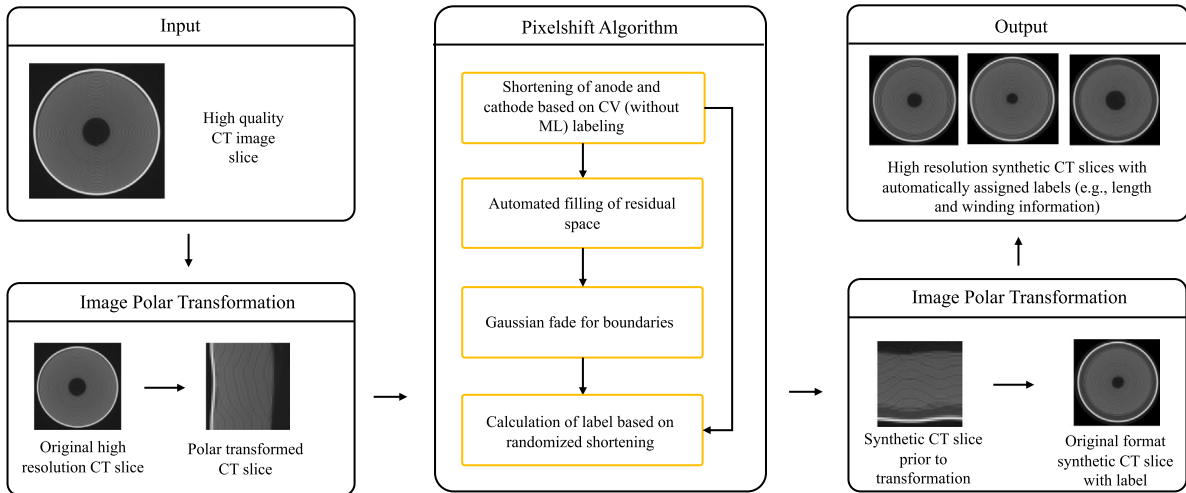


Fig. 5. CT image slice generation by pixelshift method for training of CNN based measurement algorithms of radial CT slice. High resolution CT slices serve as input to generate synthetic CT slices and automate the process of labeling. In addition to the axial CT slice a transformation of the image is performed.

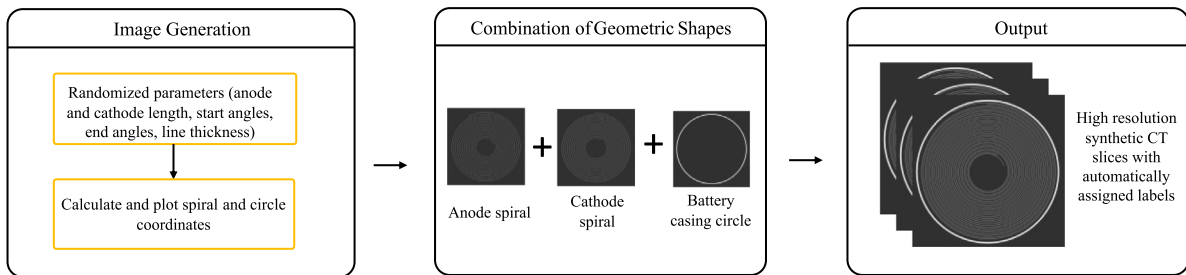


Fig. 6. CT image slice generation is achieved by combining geometric shapes from a radial CT slice. By using spirals that function as the anode and cathode together with a circle, a CT slice is synthetically created.

is transformed as shown in Fig. 5. After generating a new synthetic image and label set, a second transform is performed to return the image to its initial format. In contrast to the label in Fig. 4 (which contains a box around anode and cathode tips), the calculated label in Fig. 5 contains values of anode and cathode length and windings. An alternative method for generating synthetic radial CT slices involves combining geometric shapes, as illustrated in Fig. 6. A spiral is created for both the anode and cathode and an outer circle is added to simulate an authentic CT slice. A large dataset can be generated by varying

parameters such as anode length, cathode length, thickness, and the starting and ending angles.

Experimental

The following subchapters give an overview of the experimental procedure. This includes an overview of the cell specifications of the cylindrical cell used in this study and the specifications of the CT measurement, which is performed for method development, training,

and validation. The values are provided, to enable reproduction of the experiment.

Cell specifications

This study uses a cylindrical cell in 18650 format by manufacturer A123. The cell has two cathode tabs and one anode tab, which are welded to either the top or bottom of the cell case. The cell weighs 38.98 g. Fig. 3 shows the cell's dimensions.

Computed tomography measurements

As part of the study, a composite scan was assessed to investigate approaches towards the automation of CT-based inspection. The CT measurement is performed with a Metrotom 1500/225 kV CT scanner by company Zeiss. The voxel cloud with a resolution of $17.1\ \mu\text{m}$ is reconstructed based on 2050 projections, with 1992.82 ms integration time per projection. The tube voltage of the scan is 165 kV. The tube current is $112\ \mu\text{A}$. The image stack used for the study is generated with myVGL 3.5 software after CT imaging. The software is also used for manual inspection for validation. For the validation, the software's internal measurement tools are used.

Results

The results are presented for two representative CT slices: radial (Fig. 3(b)) and axial CT slice (Fig. 3(c)) that offer an insight into specific features and corresponding inhomogeneities. Table 1 shows an overview of the investigated inhomogeneities and the steps used in the feature extraction workflow. The workflow and results are presented for classical CV and CNN-based approaches.

Feature extraction for axial computed tomography slice

The approaches aim to identify the anode and cathode and calculate the following features: anode width, overhang, and cathode width (Table 1).

Classical computer vision-based approach

The CV-based approach aims to identify the anode first. This allows the following segmentation of the cathode and the anode overhang. Segmentation of the anode is feasible due to the high X-ray attenuation of the anode substrate. As a result anode areas are characterized by a high gray value, allowing the differentiation from surrounding cell components. Pre-processing operations such as Gaussian filtering and contrast adjustments are applied before threshold-based binarization. Due to overlapping gray values of the case and anode (Fig. 7(a)), the resulting binary image has to undergo further processing to enable the separation of the anode areas.

The cathode is identified by creating a mask around the cell's jelly roll, which is applied to the pre-processed image. Anode areas are subtracted from the prior masked image to facilitate the threshold-based binarization for segmentation.

To enable the separation of the anode overhang, a mask is created around segmented cathode areas. These areas are consequently subtracted from the binary image used to identify the anode, allowing segmentation of the overhang (Fig. 7(b)). MATLAB functions *bwconncomp()* in combination with *regionprops()* are used for feature calculation. These functions allow calculation of the anode width, cathode width, and overhang by application of bounding boxes around the areas of interest (Fig. 7(c)). The features' calculation considers anode and cathode kinks by calculating the distance between the bottom left and upper right corners of the bounding box.

Convolutional neural network-based approach

The CNN-based approach is based on the YOLOv5 network. As a baseline, this model has a high capability of fast and reliable object detection in images. Specifically, the smallest available model YOLOv5n is used in this study. As part of the YOLOv5 framework the model shows a high maturity and is widely adopted for a variety of detection applications. It also offers a faster inference time compared with newer versions such as YOLOv8n, YOLOv10n and YOLO11n for detection on the Common Objects in Context (COCO) dataset [59], which is an often used dataset for comparison of inference and accuracy of ML models [60–62].

Due to the possible different angles and shapes of the anode's overhang, simple rectangular bounding boxes around the anode overhang are not suitable for this problem. Instead, a YOLOv5n model is trained to detect the tips of anode and cathode. This model is referred to as tip detection network (TDN) in the following. It uses bounding boxes around the top and bottom tips of the anode and cathode (see Fig. 8(c)). This enables the extraction of the most crucial key points and the automated calculation of the anode and cathode width and the overhang length. Further improvement of the training and detection is achieved by cutting the CT image into four square parts and detecting each corner independently. The results are combined again, and the obtained detections for the whole image are shown (Fig. 8). As shown in the image, the algorithm accurately detects the anode and cathode tips.

For a generation of synthetic training data, we used the CT images of a single CT scan and manipulated the length and position of the anodes and cathodes. The initial images are labeled by hand, with boxes around the tip of the anode and the tip of the cathode on the top and bottom of the battery. The labels are automatically moved to match the new position as the images are manipulated. Over 16 000 new labeled images are created. Additional augmentations during training were specially selected for a more robust and reliable detection system. The image's rotation is one of the most important factors in strengthening the network for the detection of bent anodes.

Feature extraction for radial computed tomography slice

As summarized in Table 1, a total number of 4 features are extracted from the radial CT slice, the number of anode and cathode windings, and the length of the anode and cathode.

Classical computer vision-based approach

The segmentation order and principle are similar to the CV-based approach for the axial CT slice. The segmentation and identification of the anode are performed prior to the identification and measurement of cathode areas. However, before segmentation of the image and measurement, the image undergoes polar transformation (Fig. 9(a)) to enable the application of a threshold-based binarization (Fig. 9(b)) for segmentation of single windings. Fig. 9(c) shows a result for identifying a single cathode winding.

Convolutional neural network-based approach

This chapter discusses how machine learning techniques are applied to automate the calculation of the quantity of cathode and anode windings within a radial cross-section of a CT scan. ML models such as CNNs require substantial data to enhance their performance and generate more precise predictions. Given the limited available images due to the scarcity of CT images, data augmentation becomes imperative to expand the dataset and mitigate overfitting. In this process, the provided images underwent various transformations, including resizing to both larger and smaller dimensions, flipping, and rotation. Each original image was subjected to these transformations, resulting in five augmented images per original. Consequently, the dataset expanded to encompass a total of 942 images (Fig. 10(c)).

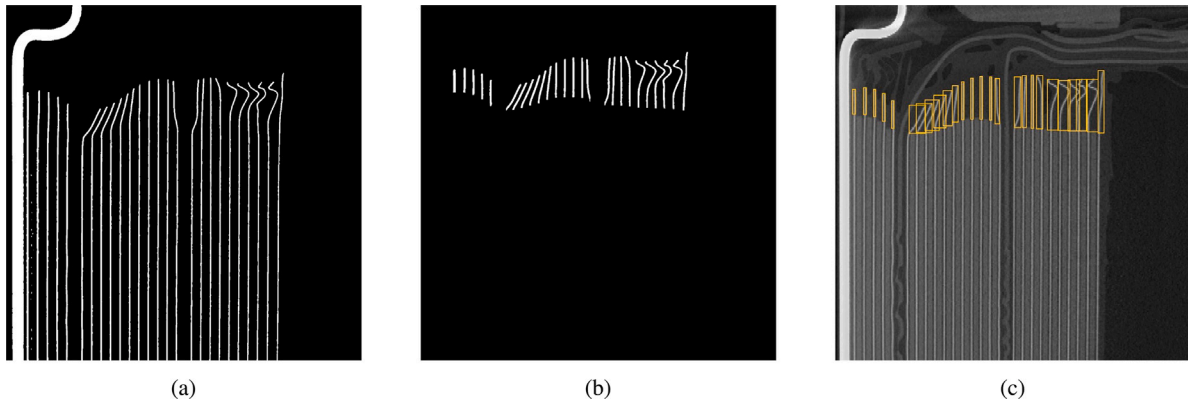


Fig. 7. Classical CV approach for axial CT slice: Selected workflow results for identification of the anode overhang showing (a) the binarized image and (b) the processed binarized image for algorithm application and detection (c).

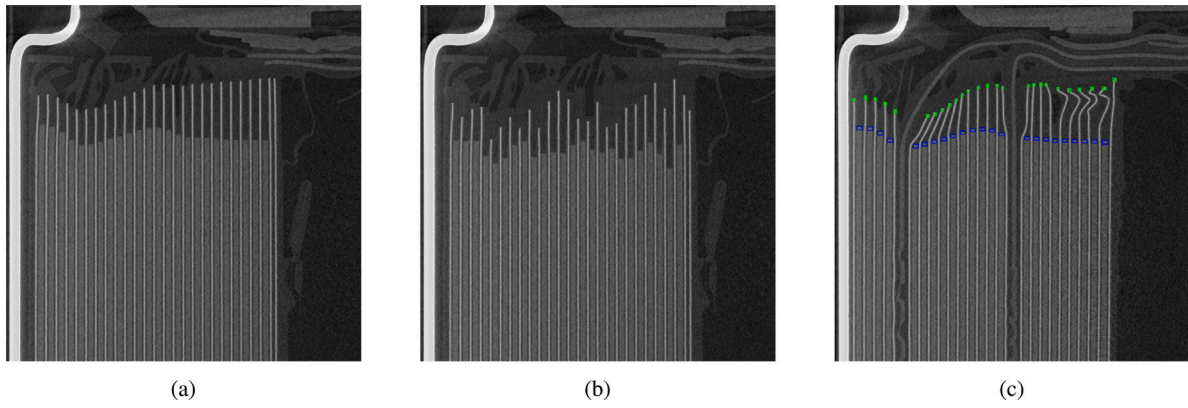


Fig. 8. CNN-based approach for axial CT slice: Three stages from training to detection for the tip detection network (TDN). The original CT slice (a) and a modified version (b) are shown, with a new validation image and its detections (c).

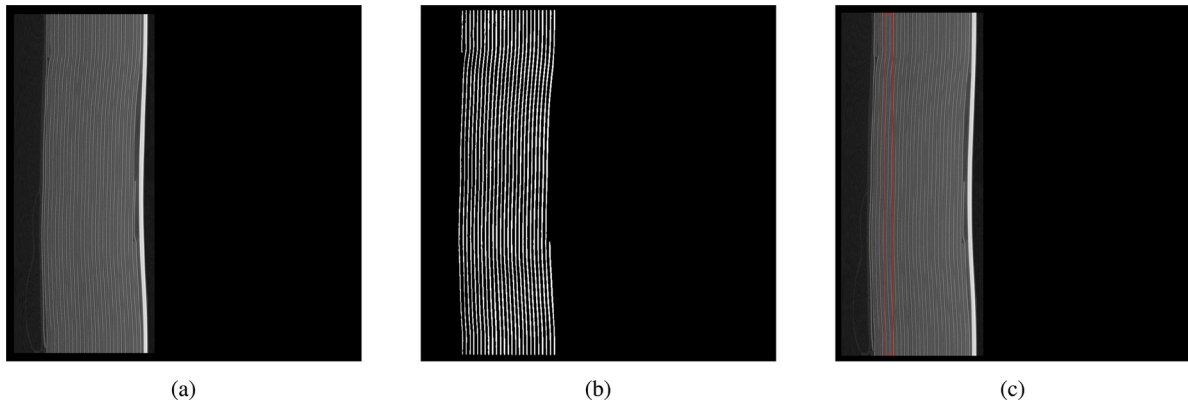


Fig. 9. Classical CV approach for radial CT slice: Selected workflow results showing a polar transform of the cell in (a) a binarized image of the polar transform (b) and an identified winding in (c).

All the provided images share a uniform number of windings, posing a challenge for training a model effectively. A synthetic dataset was generated using a Python script to address this limitation and diversify the dataset. This script facilitates the creation of images resembling the originals but with variations in windings, starting and ending angles, and the thickness of both the anode and cathode. This augmentation broadens the data frame, enhancing the model's generalization ability across a broader range of scenarios (Fig. 10(a)). The regression-based approach aims at a model capable of predicting the number of windings present in an image, even when encountering unseen images with varying winding counts from those in the training dataset. To achieve

this, a combination of synthetic data, augmented data, and original images was utilized to train various model approaches. Three distinct modeling strategies were explored. The initial modeling approach utilizes the raw images directly without any pre-processing steps. The second strategy incorporates histogram equalization to enhance the image contrast before feeding them into the model (Fig. 10(b)). A third strategy uses normalizing the images to scale the pixel values. Each of these strategies employs a unified model structure. A CNN architecture was chosen to extract features from the images. These features are then used for a regression task via convolutional, pooling, and dense layers, as detailed in Table 2.

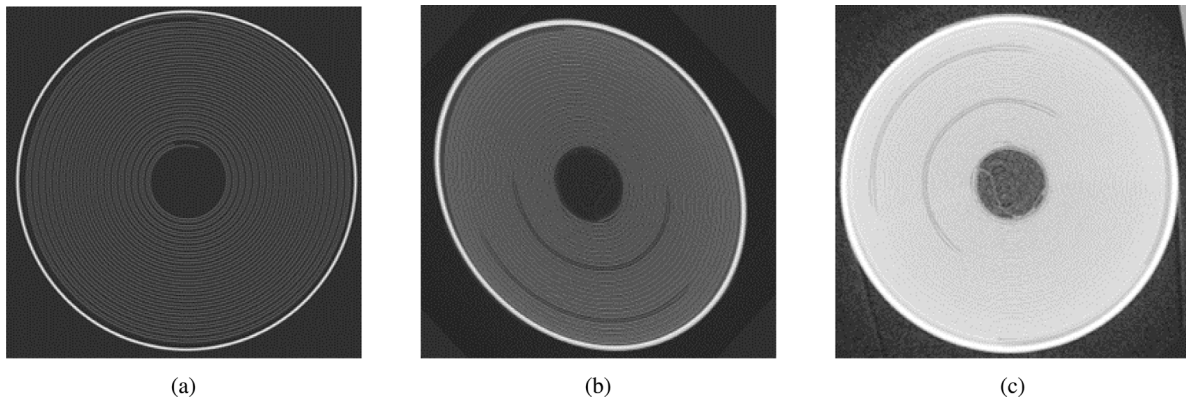


Fig. 10. CNN-based approach for radial CT slice: Synthetic data (a), data augmentation (b) and histogram equalization (c) of CT images.

Table 2

CNN-based approach for radial CT slice: CNN model architecture used for feature extraction.

Layer (type)	Configuration
InputLayer	Shape: (800, 800, 1)
Conv2D	Filters: 32, Kernel size: (3 × 3), Activation: ReLU
MaxPooling2D	Pool size: (2, 2)
Conv2D	Filters: 64, Kernel size: (3 × 3), Activation: ReLU
MaxPooling2D	Pool size: (2, 2)
Flatten	–
Dense	Units: 64, Activation: ReLU
Dense	Units: 1

To evaluate performance, the model is subjected to the entire dataset post-training to calculate the RMSE, ensuring a comprehensive assessment of its predictive accuracy (Table 3). The model that includes the normalization step demonstrates superior performance in extracting features. The approach with the pre-processing via histogram equalization was slightly better than without pre-processing but less effective than the approach with the normalization step. The RMSE results of the approaches without a pre-processing step and with normalization are displayed in Table 3 as well as the training data average.

Comparison of computer vision and convolutional neural network-based approaches

A radial and an axial CT slice are selected to compare the approaches. The slices have not previously been used to develop the CV and CNN-based measurement algorithms. Fig. 11 compares the obtained values with manual validation measurements performed in the application myVGL viewer for visualization of CT data.

The validation results indicate a good fit of both measurements based on CV and CNN detection for the axial CT slice. This is also captured by the low RMSE in Table 4. For the axial CT measurements the maximum RMSE of all measured features via CV is below 70 μm and via CNN is below 40 μm slightly above double the scan resolution of 17.1 μm . However, in Fig. 11(c), both the CNN and CV-based approaches show a deviation of the measurement values for selected middle anodes. The width of the corresponding anodes, which show a high degree of curvature in the overhang area (e.g., visible in Fig. 7(c)), are underestimated due to the chosen algorithm based on bounding

boxes. In Fig. 11(d), the CV-based approach shows a slight deviation compared to the validation measurements which is based on a variation of gray values of anode structures in outer and inner regions of the CT image of the cell.

For the radial slice, the number of windings and corresponding length of the anode and cathode are evaluated based on a single measurement value in Table 4. However, the method used in the CV approach can also evaluate each windings' length (Fig. 11(a) and Fig. 11(b)). The calculation of each winding provides an additional benefit of the CV-based approach over the CNN-based regression approach. For the radial CT, the winding and length estimation shows a lower error for the CV-based method. The maximum error of all feature measurements is 0.41% for the cathode windings compared with the validation measurement. For the CNN-based measurements, the best CNN approach is evaluated for each feature. The maximum error (anode and cathode winding estimation) for the winding estimation is 1.55%. However, the maximum error for the anode and cathode length estimation is 7.22%.

Discussion

The study results show that feature extraction can be automated for cylindrical cells with small diameters, such as those of the 18 650 format. The high resolution allows the application of classical CV and ML-based techniques to identify, extract, and measure battery cell features. However, synthetic training data sets must be generated due to the scarcity of CT images. This study presents innovative approaches for training data generation based on a single CT scan and artificially generated images. CV-based methods are compared with two CNN approaches a regression and a object detection approach.

The CV-based approach performs well for axial and radial CT slices. However, the CNN TDN approach surpassed the accuracy of the CV approach for the axial CT slice. Due to a variation of gray value between the center and outer regions of the cell, the fixed threshold in the CV approach leads to an overestimation of the cathode length in outer and an underestimation of the cathode length in inner regions. Both measurement methods lead to inaccuracies in the case of severely bent anodes. Both measurement methods could be applied in serial production environments with slight improvements.

For the radial CT slice, the CV-based approach surpassed the accuracy of the CNN regression approach. The regression approach would

Table 3

Comparison of post-training measurement results of the CNN approaches for radial CT slice.

Component feature	Regression RMSE	Normalization RMSE	Training data average
Anode windings	2.71	0.26	26.5
Anode length in mm	135.33	37.39	998.32
Cathode windings	2.35	0.30	24.5
Cathode length in mm	16.68	10.95	857.58

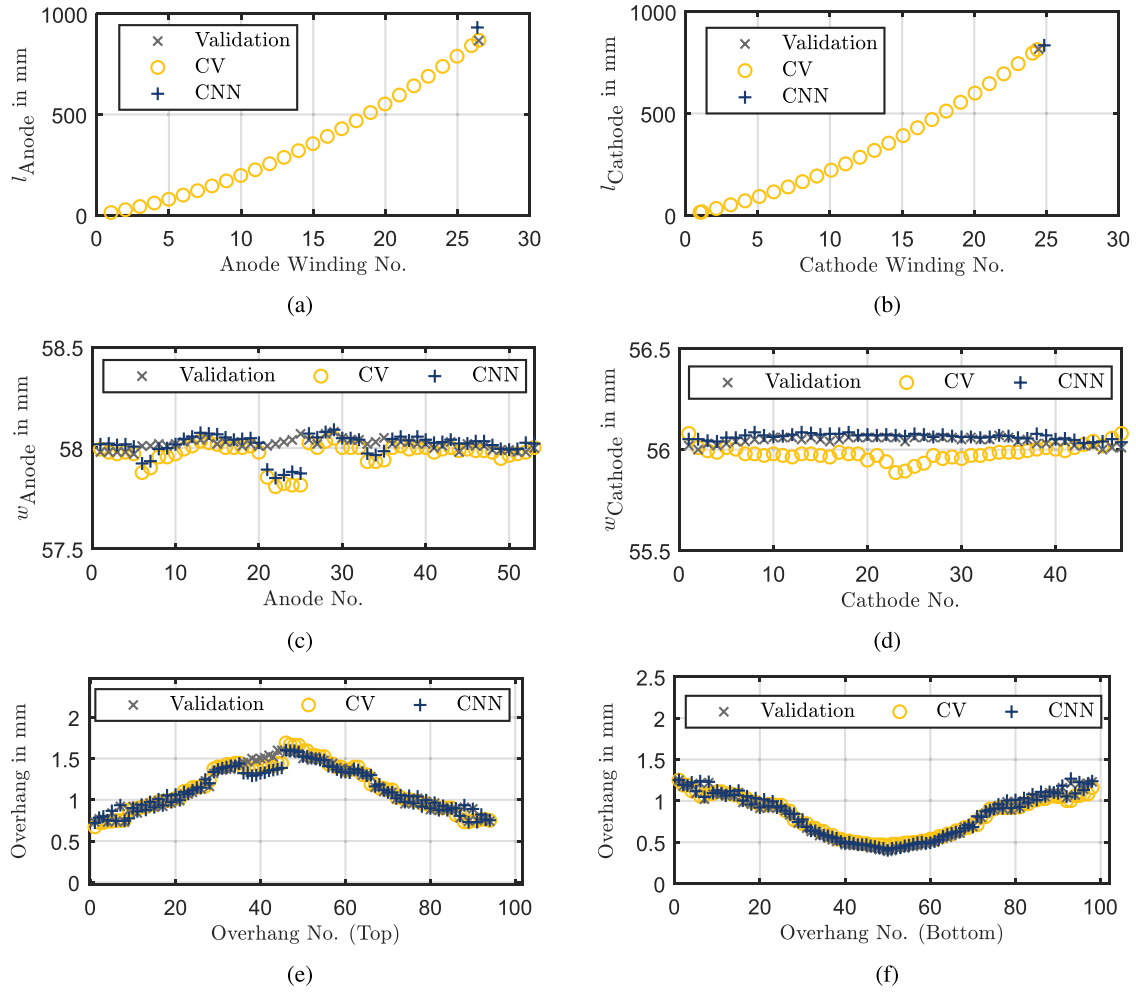


Fig. 11. Comparison of CV and CNN-based measurements. In (a) anode windings and length and (b) cathode windings and length are displayed for the radial CT slice. The results for the axial CT slice are presented in (c) for the anode and (d) for the cathode width, as well as for (e) the top and (f) the bottom anode overhang.

Table 4

Comparison of measurement results of CV and CNN approach for axial CT slice (average and RMSE values) and for radial CT slice (single value).

CT slice feature	CV	CNN	Validation
Axial CT slice			
Anode overhang in mm	0.98 ± 0.04	0.98 ± 0.03	0.97
Anode overhang (top) mm	1.14 ± 0.05	1.13 ± 0.04	1.14
Anode overhang (bot) mm	0.82 ± 0.04	0.83 ± 0.02	0.81
Anode width in mm	57.97 ± 0.05	58.01 ± 0.04	58.02
Cathode width in mm	55.98 ± 0.07	56.06 ± 0.02	56.04
Radial CT slice			
Anode windings	26.49	26.38	26.49
Anode length in mm	869.46	932.77	869.93
Cathode windings	24.37	24.85	24.47
Cathode length in mm	813.37	834.39	816.34

have to be further improved for serial production environments requiring a low measurement error. As an alternative to the improvement of the regression approach, a transfer of the TDN to the radial CT could be investigated due to its performance on the axial CT slice. A transfer could be achieved by applying the TDN network on the polar transform of the radial CT slice to enhance the accuracy of measuring the length and number of windings. This method would allow for the detection of the start and end points of the anode and cathode. Also, it would be interesting to explore the generation of training sets for new algorithms.

The CNN approaches handle deviations and gray values very well due to the extensive training with augmented data. These variations in the CT slice can also occur, e.g., due to different filters, battery active materials, component thicknesses, and resolutions, it is assumed that the transferability of CNN-based approaches surpass the transferability of CV-based approaches. Since larger cells pose a challenge regarding the CT slice resolution and image noise, it should be investigated if the workflows and approaches can be transferred to these scenarios. Additionally the application of ML-based image pre-processing steps could also be further explored. Moreover, such pre-processing may be beneficial even for CT images of cylindrical cells, particularly when these images are reconstructed from a reduced number of X-ray projections, acquired with shorter integration times (which can lead to increased image noise), or captured at lower resolutions.

While the literature shows the general ability to detect specific inhomogeneities, such as particles, kinks, and cuts, these defects do not appear often in commercial cells due to the advanced production processes and process supervision. This makes developing and training algorithms to detect the given inhomogeneities difficult. A combination or a selection of the approaches for the generation of training data explored in this work could provide opportunities for generating underrepresented defects, such as metal particle contamination in the case of LIBs, and provide opportunities for further research. In addition, the workflow could be applied to larger datasets such as provided by [63].

However, for future work and especially when being applied in inline and online applications the processing time of the algorithms should be further investigated. While this study highlights especially

the accuracy of the methods the measurement time should be part of future investigations.

Furthermore it would be of interest to also explore other methods for synthetic CT data generation and transfer these to the field of CT imaging of LIBs on cell level. Also a comparison of these methods would be of interest to the methods proposed in this work. A comparison of these methods could be performed by assessing the effect of dataset expansion on the measurement and detection results.

Conclusion

This work presents a comprehensive multi-step workflow to facilitate the automation of feature measurement in battery cells. Relevant literature for the workflow steps is identified and summarized. Especially in training data generation, a research gap hinders the development of CNN based algorithms at the early stages of battery cell production when significant amounts of imaging data are not yet available. To fill this gap, novel methods for training data generation and simultaneous labeling are introduced to train object detection algorithms in battery cells based on a single CT scan and slice — significantly reducing the need for time-intensive manual data labeling.

In this work, ML-based approaches are compared with traditional CV methods (without ML) for the measurement of internal battery cell features while highlighting the strengths of both approaches. For the axial CT slice, the TDN approach outperforms the traditional CV based approach. The highest RMSE of all measured features via CV is below 70 μm and via CNN is below 40 μm slightly above double the scan resolution of 17.1 μm .

In contrast, for the radial CT slice, the traditional CV approach outperforms the CNN regression approach, which would have to undergo significant improvement for application in a real production environment. For the radial CT, the winding and length estimation shows a lower error for the CV-based method. The maximum error of all feature measurements is 0.41% whereas the largest error for the CNN-based method is 7.22%.

The CV based approach for the axial CT slice and radial CT slice could potentially be used in serial production environments with improvements regarding the accuracy and the ability to handle contrast deviations. The CNN TDN based approach for the axial CT slice performs well if gray value and contrast deviations occur. Due to the accuracy of the TDN approach, the authors of this study see a high potential of the method to be both applied in serial production environments and undergo further development and transfer to additional chemistries and other geometries, e.g., characterized by lower resolution and image quality. The accuracy of TDN also demonstrates the potential of the proposed workflow.

CRediT authorship contribution statement

Daniel Evans: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Simon Beckmann:** Writing – review & editing, Writing – original draft, Validation, Software, Investigation. **Kevin Talits:** Writing – review & editing, Writing – original draft, Validation, Software, Investigation. **Claas Tebruegge:** Writing – review & editing, Supervision, Conceptualization. **Julia Kowal:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank HELLA GmbH & Co. KGaA and all colleagues involved for their support.

Data availability

Data will be made available on request.

References

- [1] K. Rumpf, M. Naumann, A. Jossen, J. Energy Storage 14 (2017) 224–243, <http://dx.doi.org/10.1016/j.est.2017.09.010>.
- [2] S.F. Schuster, M.J. Brand, P. Berg, M. Gleissenberger, A. Jossen, J. Power Sources 297 (2015) 242–251, <http://dx.doi.org/10.1016/j.jpowsour.2015.08.001>.
- [3] M. Dubarry, N. Vuillaume, B.Y. Liaw, Int. J. Energy Res. 34 (2) (2010) 216–231, <http://dx.doi.org/10.1002/er.1668>.
- [4] T. R  ther, M. Schamel, C. Plank, F. Schomburg, F. R  der, M.A. Danzer, J. Power Sources 587 (2023) 233677, <http://dx.doi.org/10.1016/j.jpowsour.2023.233677>.
- [5] M. Schindler, J. Sturm, S. Ludwig, J. Schmitt, A. Jossen, eTransportation 8 (2021) 100102, <http://dx.doi.org/10.1016/j.etrans.2020.100102>.
- [6] F. An, L. Chen, J. Huang, J. Zhang, P. Li, Sci. Rep. 6 (1) (2016) 35051, <http://dx.doi.org/10.1038/srep35051>.
- [7] L. Wildfeuer, M. Lienkamp, eTransportation 9 (2021) 100129, <http://dx.doi.org/10.1016/j.etrans.2021.100129>.
- [8] J.V. Barreras, T. Raj, D.A. Howey, E. Schaltz, 2017 IEEE Vehicle Power and Propulsion Conference, VPPC, 2017, pp. 1–6, <http://dx.doi.org/10.1109/VPPC.2017.8331054>.
- [9] I. Zilberman, S. Ludwig, A. Jossen, J. Energy Storage 26 (2019) 100900, <http://dx.doi.org/10.1016/j.est.2019.100900>.
- [10] D. Evans, D.M. Brieske, C. Tebruegge, J. Kowal, J. Power Sources 614 (2024) 235001, <http://dx.doi.org/10.1016/j.jpowsour.2024.235001>.
- [11] M. Ank, T. Kr  ger, M. Schreiber, M. Lienkamp, J. Energy Storage 66 (2023) 107430, <http://dx.doi.org/10.1016/j.est.2023.107430>.
- [12] J. Schnell, G. Reinhardt, Procedia CIRP 57 (2016) 568–573, <http://dx.doi.org/10.1016/j.procir.2016.11.098>, Factories of the Future in the digital environment - Proceedings of the 49th CIRP Conference on Manufacturing Systems.
- [13] S. Thiede, A. Turetskyy, A. Kwade, S. Kara, C. Herrmann, CIRP Ann 68 (1) (2019) 463–466, <http://dx.doi.org/10.1016/j.cirp.2019.04.066>.
- [14] A. Turetskyy, S. Thiede, M. Thomitzek, N. Von Drachenfels, T. Pape, C. Herrmann, Energy Technol. 8 (2) (2020) 1900136, <http://dx.doi.org/10.1002/ente.201900136>.
- [15] W. Zuo, R. Liu, J. Cai, Y. Hu, M. Almazrouei, X. Liu, T. Cui, X. Jia, E. Apodaca, J. Alami, Z. Chen, T. Li, W. Xu, X. Xiao, D. Parkinson, Y. Yang, G.-L. Xu, K. Amine, Chem. Rev. 125 (1) (2025) 369–444, <http://dx.doi.org/10.1021/acs.chemrev.4c00566>.
- [16] D. Evans, P.-M. Luc, C. Tebruegge, J. Kowal, Energies 16 (19) (2023) <http://dx.doi.org/10.3390/en16196958>.
- [17] H. Villarraga-G  mez, D.L. Begun, P. Bhattad, K. Mo, M. Norouzi Rad, R.T. White, S.T. Kelly, Nondestruct. Test. Eval. 37 (5) (2022) 519–535, <http://dx.doi.org/10.1080/10589759.2022.2070165>.
- [18] Y. Wu, S. Saxena, Y. Xing, Y. Wang, C. Li, W.K. Yung, M. Pecht, Energies 11 (4) (2018) 925, <http://dx.doi.org/10.3390/en11040925>.
- [19] P. Blazek, P. Westenberger, S. Erker, A. Brinek, T. Zikmund, D. Rettenwander, N.P. Wagner, J. Keckes, J. Kaiser, T. Kazda, P. Vyroubal, M. Macak, J. Todt, J. Energy Storage 52 (2022) 104563, <http://dx.doi.org/10.1016/j.est.2022.104563>.
- [20] S. Masuch, P. G  mbel, N. Kaden, K. Dr  der, Processes 11 (1) (2022) 10, <http://dx.doi.org/10.3390/pr11010010>.
- [21] M.D.R. Kok, A. Jnawali, T.M.M. Heenan, T.G. Tranter, D.J. Brett, P.R. Shearing, J.B. Robinson, J. Electrochem. Soc. 170 (9) (2023) 090502, <http://dx.doi.org/10.1149/1945-7111/acf412>.
- [22] T.M. Heenan, A. Jnawali, M. Kok, T.G. Tranter, C. Tan, A. Dimitrijevic, R. Jervis, D. Brett, P. Shearing, J. Electrochem. Soc. 167 (14) (2020) 140530, <http://dx.doi.org/10.1149/1945-7111/abc4c1>.
- [23] J.B. Quinn, T. Waldmann, K. Richter, M. Kasper, M. Wohlfahrt-Mehrens, J. Electrochem. Soc. 165 (14) (2018) A3284, <http://dx.doi.org/10.1149/2.028181jes>.
- [24] Z. Cai, S. Mendoza, J. Goodman, J. McGann, B. Han, H. Sanchez, R. Spray, J. Electrochem. Soc. 167 (16) (2020) 160515, <http://dx.doi.org/10.1149/1945-7111/abcabc>.
- [25] M.D. Kok, J.B. Robinson, J.S. Weaving, A. Jnawali, M. Pham, F. Iacoviello, D.J. Brett, P.R. Shearing, Sustain. Energy Fuels 3 (11) (2019) 2972–2976, <http://dx.doi.org/10.1039/C9SE00500E>.
- [26] S. Hu, J. Xu, M. Lv, Z. Zhu, J. Jia, W. Li, W. Weng, J. Phys.: Conf. Ser., Vol. 2083, IOP Publishing, 2021, 032075, <http://dx.doi.org/10.1088/1742-6596/2083/3/032075>.

- [27] X.-Y. Yao, S. Saxena, L. Su, M.G. Pecht, IEEE Access 7 (2019) 45978–45982, <http://dx.doi.org/10.1109/ACCESS.2019.2909018>.
- [28] S. Masuch, S. Gräfnitz, K. Dröder, in: K. Dröder, T. Vietor (Eds.), *Circularity Days 2024*, Springer Fachmedien Wiesbaden, Wiesbaden, 2025, pp. 53–66, http://dx.doi.org/10.1007/978-3-658-45889-8_5.
- [29] J. Wilhelm, S. Seidlmayer, P. Keil, J. Schuster, A. Kriele, R. Gilles, A. Jossen, J. Power Sources 365 (2017) 327–338, <http://dx.doi.org/10.1016/j.jpowsour.2017.08.090>, URL <https://www.sciencedirect.com/science/article/pii/S0378775317311230>.
- [30] G. Lanza, B. Haefner, L. Schild, D. Berger, N. Eschner, R. Wagner, M. Zaiß, in: W. Gao (Ed.), *Metrology*, Springer Singapore, Singapore, 2019, pp. 399–433, http://dx.doi.org/10.1007/978-981-10-4938-5_14.
- [31] A. Gabiger-Rose, *Beschleunigung der Inline-Computertomographie durch Rauschfilterung von Projektionen* (Ph.D. thesis, Dissertation), Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, 2023.
- [32] T. Dreier, D. Nilsson, E. Espes, e-J. Nondestruct. Test. (2024) <http://dx.doi.org/10.58286/29240>.
- [33] M. Vania, D. Mureja, D. Lee, J. Comput. Des. Eng. 6 (2) (2019) 224–232, <http://dx.doi.org/10.1016/j.jcde.2018.05.002>.
- [34] J. Kruth, M. Bartscher, S. Carmignato, R. Schmitt, L. De Chiffre, A. Weckenmann, CIRP Ann 60 (2) (2011) 821–842, <http://dx.doi.org/10.1016/j.cirp.2011.05.006>.
- [35] H. Villarraga-Gómez, *Proc. of Digital Imaging 2016-An ASNT Topical Conference*, 2016, pp. 44–57.
- [36] L. De Chiffre, S. Carmignato, J.-P. Kruth, R. Schmitt, A. Weckenmann, CIRP Ann 63 (2) (2014) 655–677, <http://dx.doi.org/10.1016/j.cirp.2014.05.011>.
- [37] S. Baazouzi, N. Feistel, J. Wanner, I. Landwehr, A. Fill, K.P. Birke, Batteries 9 (6) (2023) <http://dx.doi.org/10.3390/batteries9060309>.
- [38] T. Waldmann, R.-G. Scurtu, K. Richter, M. Wohlfahrt-Mehrens, J. Power Sources 472 (2020) 228614, <http://dx.doi.org/10.1016/j.jpowsour.2020.228614>.
- [39] 2018.
- [40] A. Kraemer, G. Lanza, 19th WCNDT 2016 : World Conference on Non-Destructive Testing ; 13 - 17 June, Germany . Ed.: M. Purschke, in: *Berichtsband / Deutsche Gesellschaft für Zerstörungsfreie Prüfung*, vol. 158, Deutsche Gesellschaft für Zerstörungsfreie Prüfung (DGZFP), 2016, pp. 1–7.
- [41] 2017.
- [42] N. Nawreen, U. Hany, T. Islam, 2021 International Conference on Automation, Control and Mechatronics for Industry 4.0, ACMI, IEEE, 2021, pp. 1–6, <http://dx.doi.org/10.1109/ACMI53878.2021.9528297>.
- [43] I. Szabo, J. Sun, G. Feng, J. Kanfoud, T.-H. Gan, C. Selcuk, Appl. Sci. 7 (11) (2017) 1156, <http://dx.doi.org/10.3390/app7111156>.
- [44] T. Heenan, A. Wade, C. Tan, J. Parker, D. Matras, A. Leach, J. Robinson, A. Llewellyn, A. Dimitrijevic, R. Jervis, P. Quinn, D. Brett, P. Shearing, Adv. Energy Mater. 10 (2020) <http://dx.doi.org/10.1002/aenm.202002655>.
- [45] R. Szeliski, *Computer Vision: Algorithms and Applications*, Springer Nature, 2022, <http://dx.doi.org/10.1007/978-3-030-34372-9>.
- [46] P. Sahoo, S. Soltani, A. Wong, Comput. Vis. Graph. Image Process. 41 (2) (1988) 233–260, [http://dx.doi.org/10.1016/0734-189X\(88\)90022-9](http://dx.doi.org/10.1016/0734-189X(88)90022-9).
- [47] N. O'Mahony, S. Campbell, A. Carvalho, S. Harapanahalli, G.V. Hernandez, L. Krpalkova, D. Riordan, J. Walsh, *Advances in Computer Vision: Proceedings of the 2019 Computer Vision Conference (CVC)*, Volume 1 1, Springer, 2020, pp. 128–144, http://dx.doi.org/10.1007/978-3-030-17795-9_10.
- [48] J. Canny, IEEE Trans. Pattern Anal. Mach. Intell. PAMI-8 (6) (1986) 679–698, <http://dx.doi.org/10.1109/TPAMI.1986.4767851>.
- [49] Y. LeCun, L. Bottou, Y. Bengio, P. Haffner, Proc. IEEE 86 (11) (1998) 2278–2324, <http://dx.doi.org/10.1109/5.726791>.
- [50] Y. LeCun, Y. Bengio, G. Hinton, Nature 521 (7553) (2015) 436–444, <http://dx.doi.org/10.1038/nature14539>.
- [51] J. Guo, H. Du, J. Zhu, T. Yan, B. Qiu, Comput. Methods Programs Biomed. 160 (2018) 43–49, <http://dx.doi.org/10.1016/j.cmpb.2018.03.025>.
- [52] O. Badmos, A. Kopp, T. Bernthaler, G. Schneider, J. Intell. Manuf. 31 (2020) 885–897, <http://dx.doi.org/10.1007/s10845-019-01484-x>.
- [53] D. Weimer, B. Scholz-Reiter, M. Shpitalni, CIRP Ann 65 (1) (2016) 417–420, <http://dx.doi.org/10.1016/j.cirp.2016.04.072>.
- [54] K.K. Pal, K. Sudeep, 2016 IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology, RTEICT, IEEE, 2016, pp. 1778–1781, <http://dx.doi.org/10.1109/RTEICT.2016.7808140>.
- [55] S.I. Nikolenko, *Synthetic Data for Deep Learning*, vol. 174, Springer, 2021, <http://dx.doi.org/10.1007/978-3-030-75178-4>.
- [56] C. Dewi, R.-C. Chen, Y.-T. Liu, X. Jiang, K.D. Hartomo, IEEE Access 9 (2021) 97228–97242, <http://dx.doi.org/10.1109/ACCESS.2021.3094201>.
- [57] F. Sukowski, D. Suth, A. Walldra, T. Grulich, A. Jung, 12th Conference on Industrial Computed Tomography, Fürth, Germany, iCT 2023, 2023, <http://dx.doi.org/10.58286/27727>.
- [58] L. Wunsch, K. Anding, G. Polte, K. Liu, G. Notni, Acta IMEKO 12 (4) (2023) 1–5, <http://dx.doi.org/10.21014/actaimeko.v12i4.1320>.
- [59] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C.L. Zitnick, in: D. Fleet, T. Pajdla, B. Schiele, T. Tuytelaars (Eds.), *Computer Vision – ECCV 2014*, Springer International Publishing, Cham, 2014, pp. 740–755.
- [60] G. Jocher, URL <https://docs.ultralytics.com/compare/yolov5-vs-yolov8/>.
- [61] G. Jocher, URL <https://docs.ultralytics.com/compare/yolov5-vs-yolov10/>.
- [62] G. Jocher, URL <https://docs.ultralytics.com/compare/yolov5-vs-yolo11/>.
- [63] A. Condon, B. Buscarino, E. Moch, W.J. Sehnert, O. Miles, P.K. Herring, P.M. Attia, Data Brief 55 (2024) 110614, <http://dx.doi.org/10.1016/j.dib.2024.110614>.